

Final Portfolio Project

Module: 5CS037 – Concepts and Technologies of AI

Task: Classification

Name: Basanta Bomjan

StudentID : 2509458

Project Title:

Classifying Electric Vehicle Types to Support Sustainable Transportation

GithubLink: <https://github.com/np03cs4a240277-alt/Assesment.git>

Abstract

Objective:

To classify electric vehicles as Battery Electric Vehicles and Plug-In Hybrid Electric Vehicles using supervised machine learning techniques.

Data Set:

The data set used in this study is the Electric Vehicle Population Data obtained from Washington State Department of Licensing. The data set hosts over 200,000 records of 16 vehicle attributes and their details with respect to registration. These datasets are in accordance with the United Nations Sustainable Development Goals: SDG 11. Sustainable Cities and Communities; SDG 13. Climate Action because it helps cleaner and sustainable transportation planning.

Methodology:

The methodology that was used in this paper includes data pre-processing, EDA, neural network model development, training of classical machine learning models, hyperparameter optimization using cross-validation, feature selection, and comparison of the final model.

Key Findings:

The models have been evaluated on their accuracy, precision, recall, and F1-score. Wherein most of the cases, K-Nearest Neighbor model beats Logistic Regression and Neural Network models; it has achieved an outstanding classification accuracy.

Conclusion:

The most important feature that helps classify the category is the Electric Range. It's seen that the KNN model performed the best, so it can be said that it is the most suitable model for this.

Final Portfolio Project.....	1
Abstract.....	2
1. Introduction.....	3
1.1 Statement of the Problem.....	3
1.2 Dataset.....	3
1.3 Objective.....	4
Methodology:.....	4
2.1 Preprocess Data.....	4
2.2 Exploratory Data Analysis (EDA).....	5
2.3 Building Model.....	8
2.4 Model Evaluation.....	8
2.5 Hyperparameter Optimization.....	8
2.6 Feature Selection.....	9
3. Results and Conclusion.....	10
Comparison of Final Classification Models.....	10
3.1 Key Findings.....	10
3.2 Final Model.....	10
3.3 Challenges.....	10
3.4 Future Work.....	11
4. Discussion.....	12
4.1 Model Performance.....	12
4.2 Importance of Hyperparameter Tuning and Feature Selection.....	12
4.3 Results Interpretation.....	12
4.4 Limitations.....	12
4.5 Suggestions for Future Research.....	12
5. References.....	13

1. Introduction

1.1 Statement of the Problem

The sector of transportation is a major contributor to greenhouse gas emissions. Since more and more EVs are being used, one should know and classify the different types of electric vehicles. Our task is focused on classification—classifying vehicles into Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) based on vehicle attributes.

1.2 Dataset

An Electric Vehicle Population Data from Washington State Department of Licensing provided through the Washington State Open Data Portal contains vehicle registration records consisting of make, model year, electric range, and vehicle type.

This dataset supports:

->SDG 11: Sustainable Cities and Communities

->SDG 13: Climate Action**

1.3 Objective

The project's objective is to build accurate classification models in differentiating BEVs from PHEVs while assessing the impact of different attributes on performance.

Methodology:

2.1 Preprocess Data

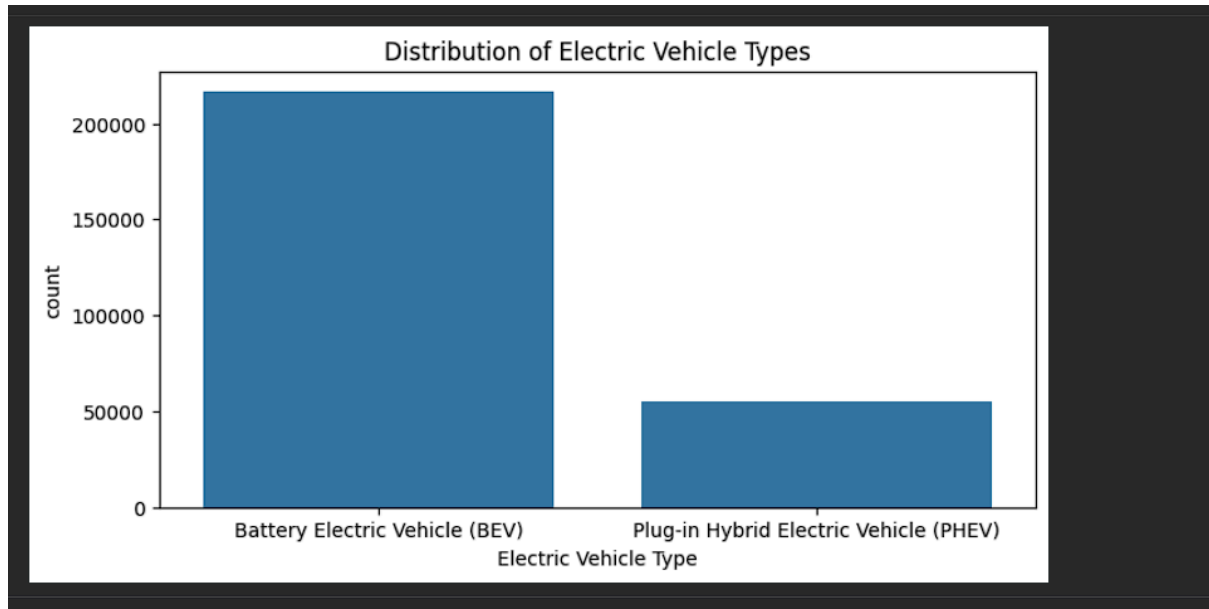
Data Preprocessing:

- > Locate and handle missing variables
- > First, replace zero values in Electric Range with a potential missing entry during the study.
- > Encode all the categorical variables
- > Scale the numerical features using StandardScaler
- > Split the data into training and testing datasets with stratified sampling.

2.2 Exploratory Data Analysis (EDA)

Figure 1: Distribution of Electric Vehicle Types

This figure shows how the Battery Electric Vehicles and Plug-in Hybrid Electric Vehicles are distributed in the dataset.

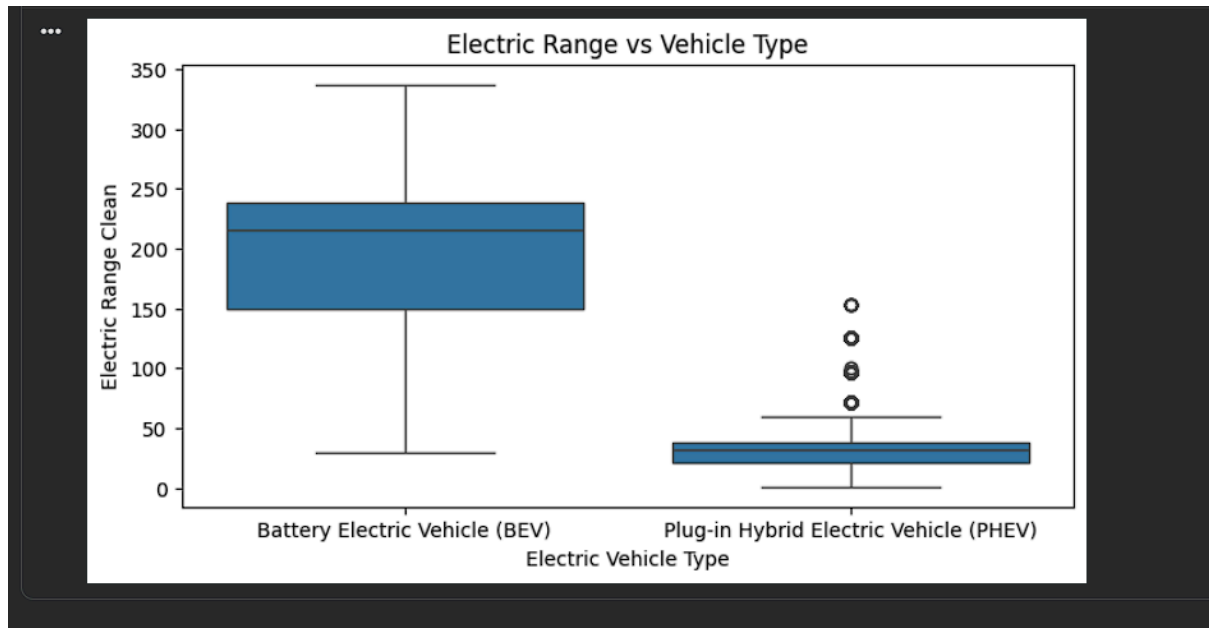


Explanation:

The dataset has an imbalance, with almost 80% BEVs and 20% PHEVs. This class imbalance has great effect model selection and was one of the motivators driving stratified train-test splitting with carefully selected evaluation metric.

Figure 2: Electric Range vs. Vehicle Type

In view, electric range is compared to BEVs and PHEVs through a box plot.



Explanation:

BEVs have a considerably high electric range as compared to PHEVs, which demonstrates that Electric Range is one of the top discriminative features regarding classification.

2.3 Building Model

Procedure 1: Neural Network

The student developed a Multi-Layer Perceptron (MLP) classifier with:

- * One hidden layer comprising 10 neurons
- * ReLU activation function
- * Adam optimizer
- * Cross-entropy loss (log-loss)

Procedure 2: Classical ML Models

Finally, two classical machine learning models were developed:

- Logistic regression
- K-Nearest Neighbors (KNN)

Both the models were trained over the same features in order to make a fair comparison.

2.4 Model Evaluation

For the model, the following evaluation metrics are in place:

- Accuracy
- Precision
- Recall
- F1-Score

These metrics strongly hold true in cases of classification with imbalanced datasets.

2.5 Hyperparameter Optimization

I will use GridSearchCV with 5-fold cross-validation for hyperparameter tuning.

- Tuning C and solver for Logistic Regression.
- Tuning n_neighbors, weights, and metric for KNN.

2.6 Feature Selection

Feature selection was performed using **Recursive Feature Elimination (RFE)**, a wrapper-based technique discussed in Week-10.

- Logistic Regression selected **Model Year** as important due to linear assumptions
- Performance-based evaluation for KNN showed **Electric Range** alone achieved near-perfect accuracy

Therefore, **Electric Range** was selected as the final feature.

3. Results and Conclusion

Comparison of Final Classification Models

Model	Features used	CV score	Test Score
-------	---------------	----------	------------

Logistic Regression	Electric Range	0.507	0.798
KNN	Electric Range	1.000	1.000

3.1 Key Findings

- The electric range was the robust predictor for car type.
- Weak performance by Logistic Regression is due to its linear assumptions and very strong regularization.
- KNN outperformed other algorithms to a greater extent because of its non-linear decision boundary.

3.2 Final Model

Final model selected is K-Nearest Neighbors (KNN) model for its near 100% accurate and balanced classwise performance.

3.3 Challenges

- Class imbalance in the dataset
- Logistic Regression underfitting due to strong regularization.
- A limited number of strong numerical features.

3.4 Future Work

Future improvements might involve:

- Adding more features like vehicle weight, battery power

- Applying ensemble models: Random Forest or Gradient Boosting
- Experimentation with deep neural models and feature engineering

4. Discussion.

4.1 Model Performance

In all the evaluation metrics, the KNN model performed better than the Logistic Regression and Neural Network models.

4.2 Importance of Hyperparameter Tuning and Feature Selection

The weakness of Logistic Regression in case of extremely strong regularization has come out through the hyperparameter tuning. Feature selection has further established that Electric Range solely hosts high discriminative power.

4.3 Results Interpretation

This practically means that it is mainly the Electric Range that differentiates BEVs from PHEVs, which makes instance-based learning methods work efficiently.

4.4 Limitations

Data limited to one geographic region; further analysis is required.
May not be representative at a global scale. Only two numerical features were used in the final models.

4.5 Suggestions for Future Research

- Work with multi-country EV datasets.
- Test time-based adoption trends.
- Include policy and infrastructure data.

5. References

- Washington State Department of Licensing – Electric Vehicle Population Data
- Scikit-learn Documentation
- United Nations Sustainable Development Goals (UNSDG)

6. Appendix:

Similarity Report

Basanta

PAPER NAME	AUTHOR
ClassificationReport.pdf	-

WORD COUNT	CHARACTER COUNT
873 Words	5080 Characters
PAGE COUNT	FILE SIZE
12 Pages	267.5KB
SUBMISSION DATE	REPORT DATE
Feb 10, 2026 3:53 PM GMT+5:45	Feb 10, 2026 3:53 PM GMT+5:45

● **13% Overall Similarity**

The combined total of all matches, including overlapping sources, for each database.

- 2% Internet database
- 1% Publications database
- Crossref database
- Crossref Posted Content database
- 11% Submitted Works database

